Question Bank: High Performance Computing (HPC)

BCSAI0304

MODULE 1: Introduction to HPC (30 Questions)

2 Marks – (K2: Understanding)

1. Define High Performance Computing (HPC).
2. What is computational thinking?
3. Mention any two applications of HPC.
4. Define cloud computing.
5. What is grid computing?
6. Define cluster computing.
7. What is quantum computing?
8. State the significance of parallel programming software platforms.
9. What is the difference between CPU and GPU?
10. Define multi-core CPU.

6 Marks – (K3: Application)

11. Explain the need and importance of High Performance Computing.
12. Compare traditional computing with high performance computing.
13. Describe the various applications of HPC in scientific research.
14. Explain how cloud computing and HPC are related.
15. Illustrate the role of GPUs in HPC environments.
16. Discuss the difference between cloud, grid, and cluster computing.
17. Explain the concept of computational thinking with an example.
18. Describe how multi-core CPUs enhance computing performance.
19. Explain the advantages of parallel programming platforms in HPC.
20. Write short notes on quantum computing and its potential.

10 Marks – (K4: Analysis)

21. Compare and contrast Cloud, Grid, Cluster, and Quantum computing.
22. Analyze the role of HPC in solving real-world scientific and engineering problems.
23. Discuss the architecture and use cases of multi-core CPUs and GPUs in HPC systems.
24. Analyze the challenges in implementing parallel programming software platforms.
25. Explain the evolution of computing and justify the need for HPC.
26. Evaluate various applications of HPC in AI and data analytics.
27. Discuss how parallel programming improves performance in modern systems.

28. Compare single-core and multi-core systems in terms of efficiency.
29. Analyze HPC's role in advancing climate modeling and simulation.
30. Discuss future trends and potential directions of High Performance Computing.

MODULE 2: Computing Architectures (30 Questions)

2 Marks – (K2)

1. Define SISD architecture.
2. Define SIMD architecture.
3. Define MISD and MIMD.
4. What is data parallelism?
5. What is task parallelism?
6. Define memory hierarchy.
7. What is concurrency?
8. Define PRAM.
9. What is NUMA?
10. Define speedup and efficiency.

6 Marks – (K3)

11. Explain Flynn's taxonomy with examples.
12. Compare SISD, SIMD, MISD, and MIMD architectures.
13. Discuss memory hierarchy in computing systems.
14. Explain bit-level and instruction-level parallelism.
15. Differentiate between concurrency and parallelism.
16. Explain decomposition and mapping with examples.
17. Discuss multithreading vs multiprocessing.
18. Explain shared memory and distributed memory models.
19. Describe the OpenMP programming model briefly.
20. Discuss speedup, efficiency, and scalability with examples.

10 Marks – (K4)

21. Compare shared memory and distributed memory models with diagrams.
22. Analyze the working of PRAM model and its variations.
23. Evaluate the impact of memory hierarchy on system performance.
24. Analyze different parallelism levels (bit, instruction, data, task) in modern CPUs.
25. Compare the performance of multithreading and multiprocessing architectures.
26. Explain in detail how OpenMP supports the shared memory model.
27. Evaluate scalability in parallel systems with examples.
28. Discuss performance metrics used in HPC systems.
29. Analyze decomposition and mapping techniques in parallel programming.

30. Discuss the challenges in designing parallel architectures.

MODULE 3: Distributed Memory (30 Questions)

2 Marks – (K2)

1. Define distributed memory.
2. What is message passing?
3. Define broadcast in message passing.
4. What is reduction operation?
5. What is scatter and gather?
6. Define parallel prefix sum.
7. What is CUDA programming?
8. Define sparse matrix.
9. Define dense matrix.
10. What is network topology?

6 Marks – (K3)

11. Explain distributed memory architecture.
12. Describe message passing networks with examples.
13. Explain the process of broadcast and reduction.
14. Discuss scatter and gather operations.
15. Explain the concept of network optimization.
16. Describe Distributed BFS using adjacency matrix.
17. Explain CUDA programming model basics.
18. Compare sparse and dense matrices.
19. Discuss matrix-based BFS algorithm.
20. Explain the importance of network topology in parallel computing.

10 Marks – (K4)

21. Analyze the working of message passing interface (MPI) in distributed systems.
22. Discuss different network topologies for parallel computing with diagrams.
23. Evaluate the performance of scatter, gather, and reduction operations.
24. Compare distributed BFS and matrix-based BFS implementations.
25. Explain CUDA programming with an example of matrix multiplication.
26. Analyze the trade-offs between sparse and dense matrix representations.
27. Discuss how network optimization improves distributed computing performance.
28. Evaluate the scalability challenges in distributed memory systems.
29. Analyze distributed graph traversal techniques.
30. Discuss the role of GPUs and CUDA in distributed HPC environments.

MODULE 4: Cluster Computing (30 Questions)

2 Marks – (K2)

1. Define cluster computing.
2. What is BLAS?
3. Define LAPACK.
4. What is clustering architecture?
5. Define clustering model.
6. What are mission-critical applications?
7. What is business-critical application?
8. Define fault detection.
9. What is checkpointing?
10. Define failover and failback.

6 Marks – (K3)

11. Explain the components of cluster computing.
12. Discuss types of clustering models.
13. Explain BLAS and LAPACK functions.
14. Compare mission-critical and business-critical applications.
15. Discuss key factors in clustering architecture.
16. Explain fault detection and masking algorithms.
17. Discuss the concept of checkpointing with example.
18. Explain heartbeat and watchdog timer mechanisms.
19. Discuss the role of fault recovery in cluster systems.
20. Explain clustering architectures with examples.

10 Marks – (K4)

21. Compare various clustering models and architectures.
22. Analyze performance optimization in cluster systems using BLAS/LAPACK.
23. Discuss fault tolerance techniques used in cluster computing.
24. Evaluate cluster computing applications in real-time systems.
25. Explain failover and failback mechanisms in fault recovery.
26. Analyze the working of checkpoint and restart mechanisms.
27. Discuss the importance of heartbeat and watchdog timers in system reliability.
28. Compare high-availability and load-balancing clusters.
29. Discuss challenges and solutions in large-scale cluster management.
30. Analyze the role of clustering in modern HPC environments.

MODULE 5: OpenMP (30 Questions)

2 Marks – (K2)

1. Define OpenMP.
2. What is the fork-join model?
3. Define a parallel region in OpenMP.
4. What are directives in OpenMP?
5. What is the purpose of runtime routines?
6. Define environment variables in OpenMP.
7. What are threads in OpenMP?
8. Define the role of compilers in OpenMP.
9. What is the goal of OpenMP?
10. List any two applications of OpenMP.

6 Marks – (K3)

11. Explain the architecture of the OpenMP API.
12. Discuss the fork-join model in detail.
13. Describe thread management and thread IDs in OpenMP.
14. Explain the parallel region construct with example.
15. Discuss the structure of a typical OpenMP program.
16. Explain how OpenMP supports shared memory parallelism.
17. Write short notes on directives and runtime routines.
18. Discuss environment variables used in OpenMP.
19. Explain how to specify the number of threads in OpenMP.
20. Discuss the advantages of OpenMP for multi-core processors.

10 Marks – (K4)

21. Analyze the workflow of an OpenMP program from fork to join.
22. Compare OpenMP with MPI in terms of architecture and usage.
23. Discuss OpenMP API components with suitable examples.
24. Evaluate OpenMP’s performance on shared memory systems.
25. Explain thread creation, synchronization, and management in OpenMP.
26. Discuss various directives and structured blocks in OpenMP programs.
27. Analyze how OpenMP compilers handle parallel regions.
28. Compare different parallel constructs available in OpenMP.
29. Discuss applications of OpenMP in real-world scientific computation.
30. Analyze the challenges and limitations of OpenMP in large-scale systems.